

MATH 3312, SPRING 2026: SOLUTIONS TO HOMEWORK 6

(1) Let V be an n -dimensional vector space, and let $T : V \rightarrow V$ be a linear transformation.

- (a) Show that any subset of V with more than n elements is linearly *dependent* (that is, it is *not* linearly independent).

Solution. Suppose that we have $n + 1$ -elements $v_1, \dots, v_{n+1} \in V$ with $n + 1$ elements: this corresponds to a linear transformation

$$\psi : k^{n+1} \xrightarrow{e_i \mapsto v_i} V.$$

By rank-nullity $\dim \ker \psi \geq (n + 1) - n = 1$, and so $\ker \psi \neq \{0\}$, which means that the vectors $v_1, \dots, v_{n+1}$ must be linearly dependent (why?).

- (b) Show that, for any $v \in V$, there exist $a_0, a_1, \dots, a_n \in k$, not all zero, such that

$$a_0 v + a_1 T(v) + a_2 T^2(v) + \dots + a_{n-1} T^{n-1}(v) + a_n T^n(v) = 0.$$

Here, T^k is simply T iterated k times.

Hint: Apply part (a) to the $n + 1$ vectors $\{v, T(v), \dots, T^n(v)\}$

Solution. As the hint says, this is a direct consequence of the fact that the $n+1$ -elements $v, T(v), \dots, T^n(v)$ are linearly dependent.

- (c) For any polynomial $p(x) \in k[x]$ with $p(x) = b_m x^m + b_{m-1} x^{m-1} + \dots + b_1 x + b_0$, show that the assignment

$$v \mapsto b_m T^m(v) + b_{m-1} T^{m-1}(v) + \dots + b_1 T(v) + b_0 v$$

is a linear transformation.

Solution. This is a direct verification.

We will denote this linear transformation by $p(T)$.

- (d) If $r(x) = p(x)q(x)$, show that $r(T) = p(T) \circ q(T)$.

Solution. You can verify this by direct computation, but here is a different perspective that will be useful later, and which we will develop soon. Let

$$\text{End}_k(V) = \{U : V \rightarrow V : U \text{ a } k\text{-linear transformation}\}.$$

We will see that this is a *ring* with the operations:

$$U_1 + U_2 : v \mapsto U_1(v) + U_2(v); \quad U_1 \cdot U_2 = U_1 \circ U_2.$$

As you know, giving a *ring* homomorphism

$$k[x] \rightarrow \text{End}_k(V)$$

amounts to specifying where the coefficients k go and also where the variable x evaluates to. Moreover, we need to ensure that wherever we evaluate x commutes with all the coefficients in k . Now, the coefficients $a \in k$ evaluate simply to the ‘scaling’ maps $v \mapsto a \cdot v$. Commuting with these is automatic by definition for any k -linear transformation. So what we are doing here is simply evaluating x at the k -linear transformation T . This *automatically* gives us a ring homomorphism $p(x) \mapsto p(T)$. In particular, it shows that $p(T) \circ q(T) = r(T)$ whenever $p(x)q(x) = r(x)$.

- (e) Show that there exists a non-zero polynomial $p(x) \in k[x]$ such that $p(T) = 0$ is the trivial linear transformation. That is,

$$p(T)(v) = 0,$$

for all $v \in V$.

Hint: Use a basis $\{v_1, \dots, v_n\}$ for V and find a polynomial such that $p(T)(v_i) = 0$ for all i .

Solution. Let $(v_1, \dots, v_n)$ be a basis for V . For each i , by part (b), there exists a non-zero polynomial $p_i(x) \in k[x]$ such that $p_i(T)(v_i) = 0$. Now take $p(x) = \prod_{i=1}^n p_i(x)$; then you can check that $p(T)(v_i) = 0$ for all i , which in turn implies that $p(T)(v) = 0$ for all $v \in V$ (why?). Note that it is important that the various linear transformations $p_i(T)$ commute with each other, but this is guaranteed by the fact that the *polynomials* $p_i(x)$ commute with each other.

Two $n \times n$ -matrices $A, B \in M_n(k)$ are **k -conjugate** if there exists an invertible matrix $S \in M_n(k)$ such that $A = SBS^{-1}$.

(2) Show that the following are equivalent for $A, B \in M_n(k)$:

- (a) A and B are k -conjugate.
- (b) There exists an n -dimensional space V with two bases $\mathcal{B}_1, \mathcal{B}_2$ and a linear transformation $T : V \rightarrow V$ such that

$$A = [T]_{\mathcal{B}_1} ; B = [T]_{\mathcal{B}_2}.$$

Solution. We showed (b) $\Rightarrow$ (a) in class: See Fact in Lecture 17. For the other implication, consider the k -linear transformation

$$T_A : k^n \rightarrow k^n$$

given by the matrix A with respect to the standard basis $\mathcal{B}_1$ for k^n . That is, we have

$$[T_A]_{\mathcal{B}_1} = A.$$

If $A = SBS^{-1}$, then by Problem 8(c) of Homework 5, we find that $\{Se_1, \dots, Se_n\}$ is also a basis $\mathcal{B}_2$ for k^n and the conjugation identity from shows that with respect to this basis we have $[T_A]_{\mathcal{B}_2} = B$.

(3) Show that the matrix

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

is $\mathbb{C}$ -conjugate but not $\mathbb{R}$ -conjugate to a diagonal matrix.

Solution. Let $k = \mathbb{C}$ or $\mathbb{R}$. Then saying that the matrix is k -conjugate to a diagonal matrix amounts to saying that there is a basis (v_1, v_2) for k^2 such that $Av_1 = \lambda_1 v_1$ and $Av_2 = \lambda_2 v_2$ for some $\lambda_1, \lambda_2 \in k$ (why? See problem 2). Now, for any vector $v = \begin{bmatrix} a \\ b \end{bmatrix}$, we have

$$Av = \begin{bmatrix} -b \\ a \end{bmatrix}.$$

Therefore, we have $Av = \lambda v$ if and only if

$$\begin{bmatrix} -b \\ a \end{bmatrix} = \lambda \begin{bmatrix} a \\ b \end{bmatrix} \Leftrightarrow -b = \lambda a ; a = \lambda b \Rightarrow a = -\lambda^2 a \Rightarrow \lambda^2 = -1.$$

Clearly, such a λ does not exist in $\mathbb{R}$, which tells us that the matrix is not $\mathbb{R}$ -conjugate to a diagonal matrix.

If we work with $\mathbb{C}$ instead, then we find that $\lambda = \pm i$ would work. In this case, the equation

$$\begin{bmatrix} -b \\ a \end{bmatrix} = \pm i \begin{bmatrix} a \\ b \end{bmatrix}$$

admits $\begin{bmatrix} 1 \\ \mp i \end{bmatrix}$ as a solution. It is easy to see that the vectors

$$\begin{bmatrix} 1 \\ -i \end{bmatrix}, \begin{bmatrix} 1 \\ i \end{bmatrix}$$

form a basis $\mathcal{B}$ for $\mathbb{C}^2$, and with respect to this basis, we have $[T_A]_{\mathcal{B}} = \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix}$, meaning that A is $\mathbb{C}$ -conjugate to a diagonal matrix.

An $n \times n$ -matrix is **upper triangular** if it doesn't have any non-zero entries below the diagonal. That is, if $a_{i,j} = 0$ whenever $i > j$.

- (4) Show that the following are equivalent for a matrix $A \in M_n(k)$ (here, T_A is the linear transformation $k^n \rightarrow k^n$ with matrix A):

- (a) A is k -conjugate to an upper triangular matrix $B = [b_{i,j}] \in M_n(k)$.
- (b) There exists a basis $\{v_1, v_2, \dots, v_n\}$ of k^n such that, for each j ,

$$T_A(v_j) = \sum_{i \leq j} b_{i,j} v_i,$$

for some $b_{i,j} \in k$.

- (c) There exists a basis $\{v_1, v_2, \dots, v_n\}$ of k^n such that, for each i ,

$$T_A(\langle v_1, \dots, v_i \rangle) \leq \langle v_1, \dots, v_i \rangle.$$

Solution. (1) $\Leftrightarrow$ (2): This is just from the fact that a matrix conjugate to A is the same as a $\mathcal{B}$ -matrix $[T_A]_{\mathcal{B}}$ for some basis $\mathcal{B}$ for k^n . Saying that such a matrix $[b_{i,j}]$ is upper triangular is the same as saying that $b_{i,j} = 0$ for $i > j$.

(2) $\Leftrightarrow$ (3): This arises from the fact that both statements are equivalent to saying that, for each j , $T_A(v_j)$ is a k -linear combination of $v_1, \dots, v_j$.

A **k -pair** is a pair (V, T) , where V is a k -vector space and $T : V \rightarrow V$ is a k -linear transformation. Suppose that we have a collection of k -pairs

$$(V_1, T_1), \dots, (V_r, T_r).$$

The **product k -pair** of such a collection, denoted

$$(V_1, T_1) \times \dots \times (V_r, T_r)$$

is the k -pair with k -vector space $V_1 \times \dots \times V_r$ and linear transformation

$$(v_1, \dots, v_r) \mapsto (T_1(v_1), \dots, T_r(v_r)).$$

- (5) Suppose that we have a collection as above and bases $\mathcal{B}_1, \dots, \mathcal{B}_r$ for $V_1, \dots, V_r$ such that $[T_i]_{\mathcal{B}_i} = A_i$. Then the product k -pair is isomorphic to (k^n, T_A) where:
- $n = \dim_k V_1 + \dots + \dim_k V_r$.
 - A is the **block diagonal matrix**

$$\begin{bmatrix} A_1 & 0 & \dots & 0 \\ 0 & A_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & A_r \end{bmatrix}$$

Solution. We have to set up an isomorphism $\psi : k^{n_1} \times \dots \times k^{n_r} \xrightarrow{\sim} k^n$ such that we have a 'commuting' rectangle

$$\begin{array}{ccc} k^{n_1} \times \dots \times k^{n_r} & \xrightarrow[\simeq]{\psi} & k^n \\ T_{A_1} \times \dots \times T_{A_r} \downarrow & & \downarrow T_A \\ k^{n_1} \times \dots \times k^{n_r} & \xrightarrow[\simeq]{\psi} & k^n \end{array}$$

Such an isomorphism ψ is obtained in the following way: It sends the basis elements e_j of k^{n_1} to e_j in k^n . More generally, it sends the basis elements e_j of k^{n_i} to $e_{n_1 + \dots + n_{i-1} + j}$ in k^n .